COMP 532 Machine Learning and Bioinspired Optimisation

Lecture 6 Reinforcement Learning

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Recap

- Reinforcement Learning
  - Multi-armed bandits
- Games
  - 10-arm bandit example

Overview

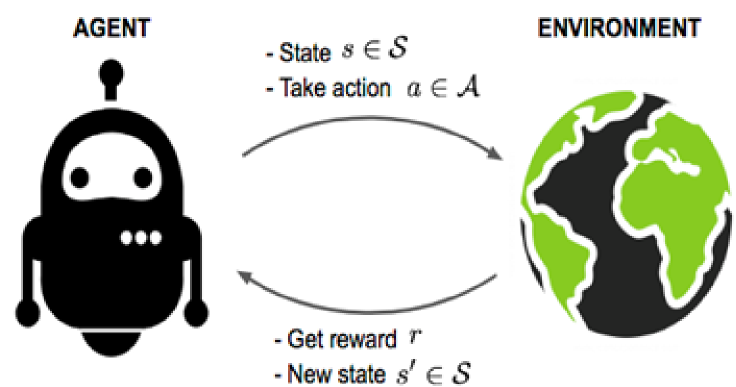
- Problem Solving from Nature
- Reinforcement Learning (RL)
  - Roots of RL
  - Key features of RL
  - Elements of RL
  - Multi-armed bandits
  - Solve the problem in Markov Decision Processes (MDPs)
- Deep Learning
- Multi-Agent Learning
- Swam Intelligence

The Agent-Environment Interface

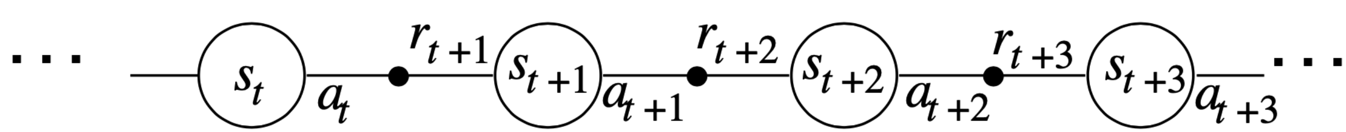
Reinforcement learning is based on interaction between an agent and an environment.

At discrete time steps $t = 0,1,2, \ldots$:

- State: $S_t \in \mathcal{S}$
- Action: $A_t \in \mathcal{A}(S_t)$
- Reward: $R_{t+1} \in \mathbb{R}$
- Next state: $S_{t+1} \in \mathcal{S}$



The agent selects actions; the environment responds with rewards and new states.



Policy

A **policy** specifies how the agent behaves.

- Stochastic policy:

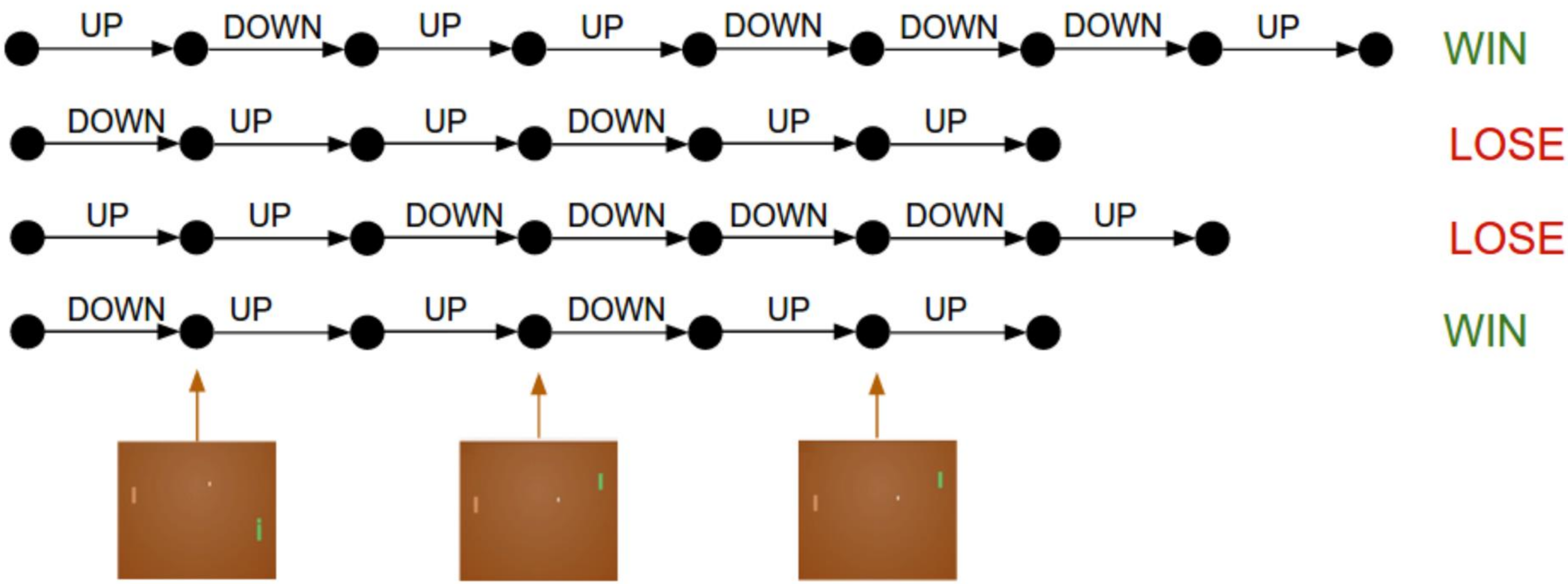
$$\pi(a \mid s) = P(A_t = a \mid S_t = s)$$

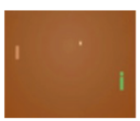
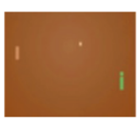
- Deterministic policy:

$$a = \pi(s)$$

- Learning in reinforcement learning means **improving the policy from experience**.

The Agent Learns a Policy



- $\pi($  , UP) = 0.8
- $\pi($  , DOWN) = 0.2

Policy: π(State, Action) => Probability

Atari game: Pong

Goals and Rewards

Reinforcement learning assumes that goals are expressed via a scalar reward signal.

Key principles:

- Rewards specify *what* we want, not *how* to achieve it
- Rewards must be observable by the agent
- Rewards may be sparse or delayed

Reward Hypothesis:

- All goals can be represented as the maximisation of expected cumulative reward.

Returns: What the Agent Optimises

The agent does not maximise immediate reward, but **return**.

Episodic Tasks

- Tasks with natural termination (e.g. games):

$$G_t = R_{t+1} + R_{t+2} + \cdots + R_T$$

Continuing Tasks

- Tasks without terminal states use discounted return:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

where $0 \leq \gamma \leq 1$is the discount factor.

The discount factor controls:

- Preference for immediate vs future rewards
- Mathematical convergence of returns

Interpretation:

- $\gamma = 0$: myopic (short-sighted)
- $\gamma \approx 1$: far-sighted

The Markov Property

- A state has the **Markov property** if it summarises all relevant past information.

Intuition:

- The future depends on the present state and action, not the full history.

Formally:

$$P(S_{t+1}, R_{t+1} \mid S_t, A_t)$$

$$P\left(s_{t+1} = s', r_{t+1} = r | s_t, a_t, r_t, s_{t-1}, a_{t-1}, r_{t-1}, \ldots, s_0, a_0, r_0\right)$$
$$= P\left(s_{t+1} = s', r_{t+1} = r | s_t, a_t\right)$$

for all s' and r, and all histories

$s_t, a_t, r_t, s_{t-1}, a_{t-1}, r_{t-1}, \ldots, s_0, a_0, r_0$

Why the Markov Property Matters

If the Markov property holds:

- Planning and learning become tractable
- Value functions can be defined recursively

If it does not hold:

- Learning becomes significantly harder
- The agent may need memory or belief states

Markov Decision Processes (MDPs)

- A reinforcement learning problem with the Markov property is a **Markov Decision Process**.
- An MDP is defined by the tuple:

$$\langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$$

Where:

  - $\mathcal{S}$: set of states
  - $\mathcal{A}$: set of actions
  - $P(s', | sa)$:transition probabilities
  - $R(s, a)$:expected reward
  - γ: discount factor

Interaction in an MDP

At each time step:

- Agent observes S_t
- Chooses $A_t \sim \pi(\cdot \mid S_t)$
- Environment samples:
  - $S_{t+1} \sim P(\cdot \mid S_t, A_t)$
  - $R_{t+1} \sim R(S_t, A_t)$
- This cycle continues until termination (or indefinitely).

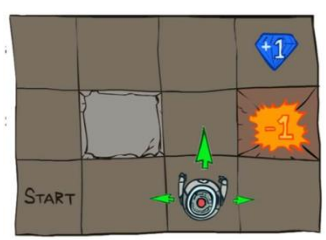
Bandits as a Special Case of MDPs

- Bandits can be viewed as MDPs with:
- A single state
- Immediate rewards
- No state transitions
- This shows how MDPs strictly generalise bandit problems.

Example: Gridworld MDP

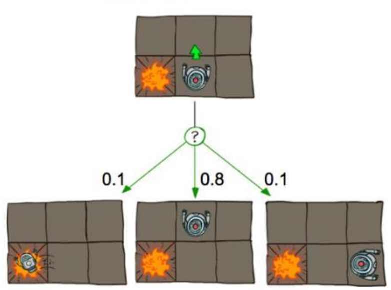
Typical gridworld specification:

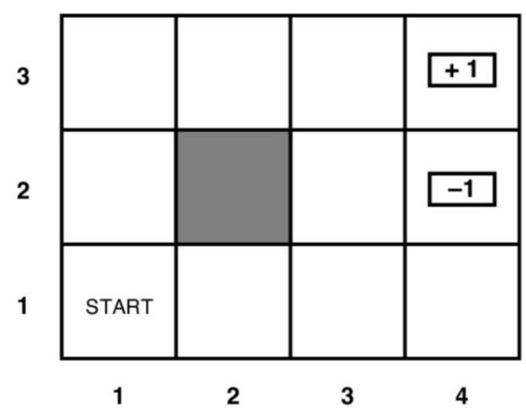
- States: grid cells
- Actions: up, down, left, right
- Transitions: stochastic movement
- Rewards: small negative per step, terminal rewards
- Used to illustrate planning and learning algorithms.



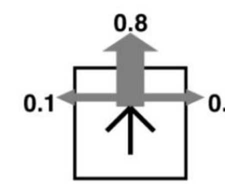
R(s) = -0.04 for every non-terminal state

Transition model:





Transition model:



R(s) = -0.04 for every non-terminal state

Summary

In this lecture, we:

- Extended bandits to sequential decision problems
- Introduced agent–environment interaction
- Defined returns and discounting
- Formalised the Markov property
- Defined Markov Decision Processes

MDPs are the foundation for value functions and Bellman equations.

Home task: Read Chapter 3 of the Reinforcement Learning book